Smart Classroom Energy Management System

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Abstract

- The need of the project
- Objective/ Goal
- Methodology followed
- Expected Outcome of the project
- Advantages
- Who are all the Beneficiaries
- Social Impact
- Road map to complete the project
- References (take latest 2022,2021 projects)

(these points you can discuss in different different slide)

Need of the Project

- In most classrooms, lights and fans are operated manually. Very often, they are left switched ON even when the classroom is empty or when they are not required. This leads to unnecessary power consumption and energy wastage. In large educational institutions, this small issue across many classrooms results in a significant increase in electricity bills and inefficient energy usage.
- There is a need for a simple, low-cost system that can automatically manage classroom electrical appliances based on actual usage, without depending completely on human intervention. This project addresses that need by using sensors and a microcontroller to ensure that energy is used only when required.

Objective / Goal

- The main objective of this project is to design and implement a smart system that automatically controls classroom lights and fans based on the presence of people.
- The goals of the project are:
- To reduce unnecessary energy consumption in classrooms
- To automatically switch ON lights and fans when the classroom is occupied
- To switch OFF appliances when the classroom is empty
- To provide a manual override option for teachers
- To develop a low-cost and easily deployable solution using a microcontroller

Methodology Followed

- The project works by continuously monitoring the classroom environment using sensors and controlling electrical appliances through a microcontroller.
- First, a PIR motion sensor is placed inside the classroom to detect human presence. When motion is detected, the sensor sends a signal to the microcontroller. The microcontroller processes this input and turns ON the lights and fans using relay modules.
- If no motion is detected for a predefined period of time, the microcontroller assumes that the classroom is empty and automatically turns OFF the appliances. A timer mechanism is used so that brief inactivity does not cause unnecessary switching.
- A manual override switch is also provided so that the teacher can control the appliances manually in special situations. The entire logic is implemented using embedded programming on the microcontroller.

Expected Outcome of the Project

- The expected outcome of this project is an automated classroom setup where lights and fans operate only when required. The system should successfully detect occupancy, control appliances in real time, and reduce power wastage.
- By implementing this system:
- Energy consumption in classrooms will be reduced
- Manual effort required to control appliances will be minimized
- The system will demonstrate effective use of sensors, timers, and control logic
- A working prototype suitable for real classroom environments will be achieved

Advantages

- Reduces unnecessary electricity usage
- Saves energy and lowers electricity bills
- Simple design and easy to implement
- Low cost compared to complex smart systems
- Requires minimal human intervention
- Can be easily expanded to multiple classrooms

Beneficiaries

- The main beneficiaries of this project are:
- **Students**, who get a comfortable learning environment
- **Teachers**, who no longer need to manually manage appliances
- **Educational institutions**, which benefit from reduced energy costs
- **Society**, due to improved energy conservation practices

Social Impact

- Energy conservation is an important social responsibility. This project encourages efficient use of electricity and reduces wastage in public spaces like classrooms. By adopting such systems, educational institutions can contribute toward sustainable development and promote awareness about responsible energy usage among students.
- This project also helps students understand how simple embedded systems can be used to solve real-world problems, encouraging innovation and practical learning.

Road Map to Complete the Project

Phase	Description
Week 1	Finalizing project idea and system requirements
Week 2	Procurement of components
Week 3	Circuit design and sensor testing
Week 4	Writing and testing microcontroller code
Week 5	Integration of hardware and software
Week 6	Testing, debugging, and fine-tuning
Week 7	Documentation and final demonstration

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